ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY LEVEL 5

TERM 3

UNIT:INSTALL COMPUTER SOFTWARE

UNIT CODE: **0619 451 06A**

*ELEMENT COVERED:***TEST COMPUTER SOFTWARE FUNCTONALITY**

TIME 4 HOURS

TOTAL 50 MARKS

Instructions

- ✓ *Read **all** the instructions and the scenario carefully before you begin. You have 10 minutes of reading time included in the 3 hours.*
- ✓ *This paper has **eight tasks**. Attempt **all** of them.*
- ✓ *Work **individually**. Classmates may help only in Task 5, item 6 (the usability test), where they act as test users.*
- ✓ *Before you begin, create a folder named YourName_LibreOfficeTest on the computer. Save **all** your documents, spreadsheets and screenshots in it. Work saved elsewhere will not be marked.*
- ✓ *Download LibreOffice only from the official link given on the paper: https://www.libreoffice.org/download/download-libreoffice/*
- ✓ *Use your **own words**. Content copied directly from the internet or from another candidate will not earn credit.*
- ✓ *Label every screenshot with its **Test ID** or **Task number**. Unlabelled screenshots will not be accepted as evidence.*
- ✓ *Present your documents professionally. Use clear headings, tables where required, correct spelling and a consistent font and layout.*
- ✓ *Your work will be assessed on:*
  - *accuracy and completeness of your answers*
  - *quality of your test plan and test cases*
  - *evidence of actual testing, such as screenshots and records*
  - *clarity and professionalism of your reports*
  - *sound judgement in your final recommendation*
- ✓ *Manage your time. A suggested guide is below.*

You will need:

- A computer with internet access
- LibreOffice (free download): https://www.libreoffice.org/download/download-libreoffice/
- A spreadsheet or word processor for documentation
- A screenshot tool, such as Snipping Tool, PrtScn or Snip & Sketch

Scenario

Greenfield Secondary School currently uses a paid word processor in its computer lab and administration office. The licence is expensive, so the Board of Management wants to replace it with **LibreOffice Writer**, a free and open-source word processor.

The school will use Writer for:

- Typing and formatting exam papers, circulars and reports
- Preparing students' assignments
- Producing letters to parents, including tables and school logos
- Saving documents in formats compatible with Microsoft Word and PDF

You have been hired as a **Software Tester** in the school's ICT department. The Board has asked you to test Writer and recommend whether it is ready for school-wide use.

Create a folder named YourName_LibreOfficeTest and save all your work, screenshots and documents in it.

Task 1: Prepare the Testing Environment

1. Download LibreOffice from the link above and install it on your computer.
2. Open Writer and record the test environment in a table: operating system and version, LibreOffice version, computer specifications (RAM, processor), and date of testing.
3. Take a screenshot of the **Help > About LibreOffice** window as proof of installation.

Task 2: Importance of Software Testing

The school's principal says, *"The software is already used worldwide, so there is no need to test it."*

Write a one-page memo to the principal explaining why testing is still necessary before the school adopts Writer. Your memo must:

- Give at least **four** reasons, with each linked to the school's needs. Consider data loss, document compatibility, cost of fixing problems after rollout, user training and confidence, and the accuracy of official documents.
- Explain the difference between **verification** and **validation**, using Writer as your example.
- State the likely consequences if the school adopts Writer without testing.

Task 3: Develop a Test Plan

Prepare a formal **Test Plan** for testing LibreOffice Writer. It must contain the following sections:

1. **Test plan title and identifier**
2. **Introduction and objectives:** what the testing aims to achieve
3. **Scope:** features to be tested and features not to be tested, with reasons
4. **Features under test:** choose at least **six** from the list below
    - Text entry and formatting (font, size, bold, italics, alignment, spacing)
    - Find and Replace, and Spell Check
    - Tables (insert, merge cells, add and delete rows)
    - Images and shapes (insert, resize, wrap text)
    - Page layout (margins, orientation, headers, footers, page numbers)
    - Lists, styles and Table of Contents
    - Save, Save As and Export (ODT, DOCX, PDF)
    - Opening and editing Microsoft Word (.docx) files
    - Undo/Redo and AutoRecovery
    - Printing and Print Preview
5. **Testing techniques to be used** and the reason for choosing each
6. **Testing tools to be used** (see Task 6)
7. **Test environment** (from Task 1)
8. **Roles and responsibilities**
9. **Schedule** (a simple timeline for the testing activities)
10. **Entry and exit criteria:** when testing starts and when it can be considered complete
11. **Risks and assumptions**
12. **Deliverables:** test cases, defect reports, test summary report

Task 4: Design the Test Cases

Design a test case table with **at least 6 test cases** covering the features in your test plan. Use these columns:

| Test ID | Feature | Technique | Test description and steps | Test data | Expected result | Actual result | Status (Pass/Fail) | Remarks |

Your test cases must include **normal, boundary and invalid** conditions. For example, try to insert a table with 0 rows

Task 5: Execute the Tests

1. Execute every test case you designed in Task 4 on LibreOffice Writer.
2. Fill in the **Actual result** and **Status** columns.
3. Take screenshots as evidence for **at least ten** of your test cases and label each one with its Test ID.
4. Carry out a **compatibility test**:
    - Create a document in Writer with a heading, a table, an image and a bulleted list.
    - Save it as .docx and .pdf.
    - Reopen the .docx file in Writer, or in any other word processor available to you, and record whether the formatting stayed the same.
5. Carry out a **performance test**: open or create a document with at least 50 pages (you can paste repeated text) and record how Writer behaves when scrolling, searching and saving.
6. Carry out a short **usability test**: ask two classmates to complete a simple task in Writer, such as creating a letter with a table and saving it as PDF. Record how long it took each of them, the difficulties they faced, and their suggestions.